CARDIAC



In patients 45 years or older hospitalized for at least one night after elective non-cardiac surgery, the overall 30-day mortality rate in a large multinational study involving more than 40,000 patients was 1.8%, with variations in rate depending on region of the world.

For the 22,447 patients within the study from North America, Europe and Australia, overall 30-day mortality was 1.1% .The 30-day mortality risk for non-elective urgent/emergent surgery is at least double the risk of elective surgery. [1]

Many of these deaths are linked to cardiac complications. [1] Myocardial Injury after Noncardiac Surgery (MINS) occurs in 12%-24% of these mortality cases, as indicated by troponin levels exceeding the 99th percentile within 3 days of surgery and presumed to be due to myocardial ischemia,. [2] MINS is associated with a postoperative 30-day mortality rate of 9.8%, compared to 1% for those without it, and increases the risk of major vascular complications, such as myocardial infarction and stroke [3,4,5]. Most patients suffering MINS postoperatively are asymptomatic for ischemic features or symptoms. Without postoperative troponin testing, 90% of patients experiencing MINS would remain undetected. [3, 6, 7]

Screening Tools

The Revised Cardiac Risk Index (RCRI) is a six-question tool used to estimate a patient's risk of major perioperative cardiac complications after non-cardiac surgery. [8, 9] For patients aged ≥ 65, or for patients aged 45-64 with either significant cardiovascular disease, or an RCRI score ≥ 1, measuring NT-proBNP or BNP before non-cardiac surgery can improve risk assessment [10]. When the postoperative risk of MINS is deemed to be greater than a 5% risk based on NTproBNP or BNP levels, then postoperative MINS Surveillance by way of Troponin testing is indicated [10]

Revised Cardiac Risk Index

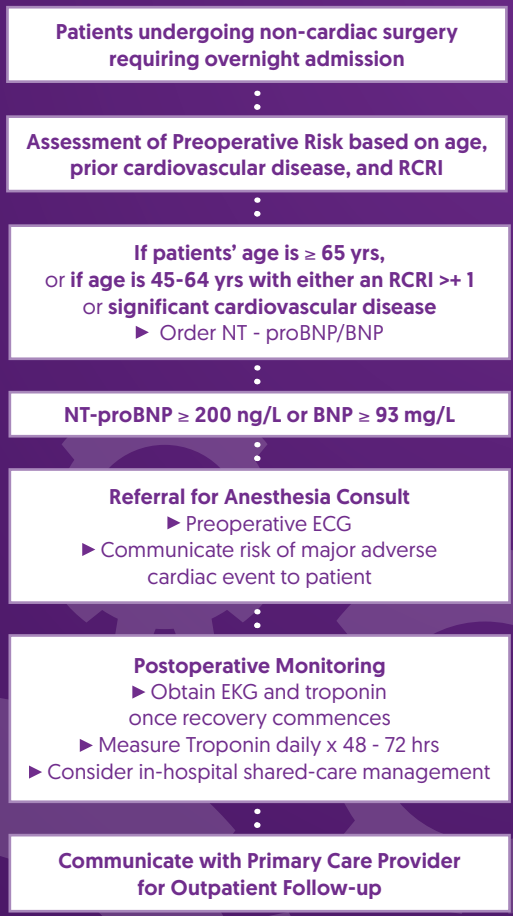
RCRI SCREENING	YES	NO
1. History of ischemic heart disease	☐	☐
2. History of congestive heart failure	☐	☐
3. History of cerebrovascular disease	☐	☐
4. Use of insulin therapy for diabetes	☐	☐
5. Preoperative serum creatinine > 177 µmol/L (>2.0 mg/dL)	☐	☐
6. High-risk surgery (Intraperitoneal, intrathoracic, or suprainguinal vascular surgery)	☐	☐

SCORING Add 1 point for every **YES**

Total RCRI Points	Risk Estimate %	95% CI for the Risk Estimate
0	3.9	2.8%-5.4%
1	6.0	4.9%-7.4%
2	10.1	8.1%-12.6%
≥3	15.0	11.1%-20.0%

Total RCRI Score and corresponding risk of myocardial infarction, cardiac arrest, or death at 30 days after noncardiac surgery.

CARDIAC ALGORITHM



Prehabilitation and Optimization Recommendations

Order NT-proBNP/BNP	⚙ Criteria: overnight admission and one of the following: age ≥ 65 OR age 45-64 with either RCRI ≥ 1 or significant cardiovascular diseasee ⚙ Threshold values are different for each test (BNP or NTproBNP) for triggering postoperative monitoring with troponin
Referral for Anesthesia Consult	⚙ Perioperative risk discussion and planning ⚙ If NT-proBNP is unexpectedly high without an obvious cause, consider further evaluation, including a physical exam and echocardiogram ⚙ Communicate risk estimates with patient based on RCRI score or elevated NT-proBNP values (8% for 200-1500 pmol/mL, and 16% for > 1500 pmol/mL, risk of death is 1.4% and 4.0% respectively) [8] *Note these risk values are overall results from large trials including different surgerytypes. Risk stratification may vary based on the type of surgery planned.*
Postoperative Monitoring	⚙ Obtain EKG and troponin on the day of surgery once recovery commences ⚙ Measure Troponin daily x 2-3 days (should not prolong hospitalization for troponin monitoring) ⚙ If Troponin > 99th percentile, MINS management required [1] ⚙ Consider in-hospital shared-care management
Communicate with Primary Care Provider for Outpatient Follow-up	⚙ Establish follow-up plan for MINS positive patients after discharge from hospital given their increased risk of postoperative mortality